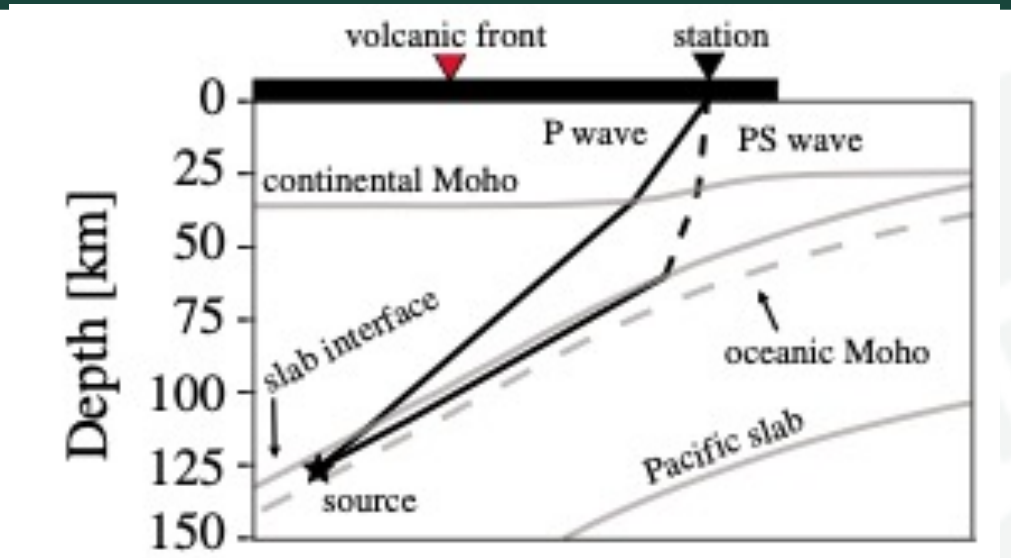
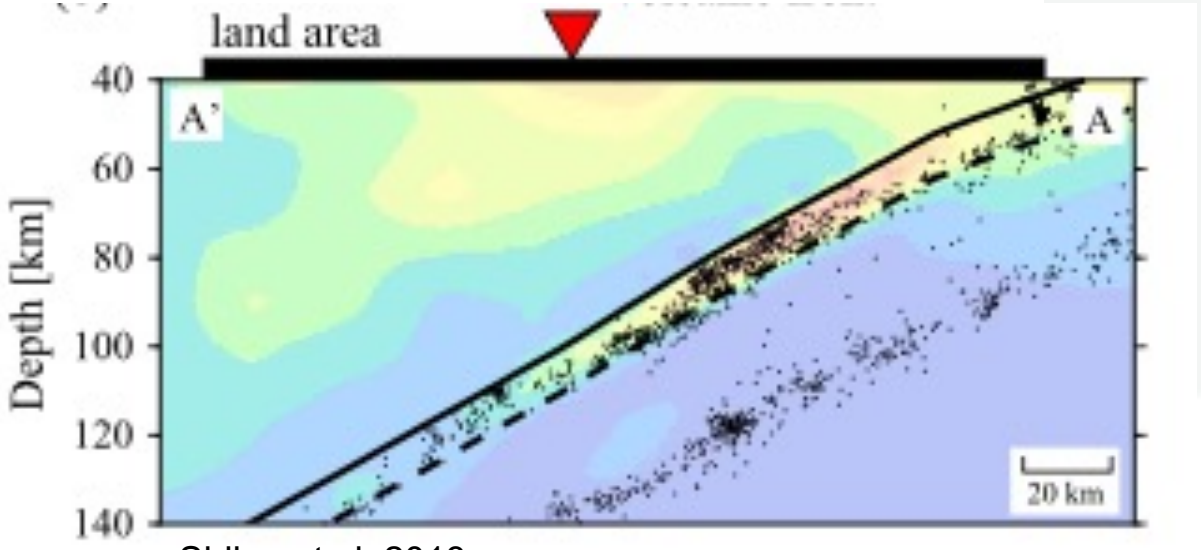
Detecting converted seismic phases of Tonga deep earthquakes: insights from deep-learning methods

Ziyi Xi[1], S. Shawn Wei[1], Nooshin Saloor[1], Weiqiang Zhu[2], Gregory C. Beroza[3]

[1]Michigan State University
[2]California Institute of Technology
[3]Stanford University

Use seismology to get the accurate slab crust position



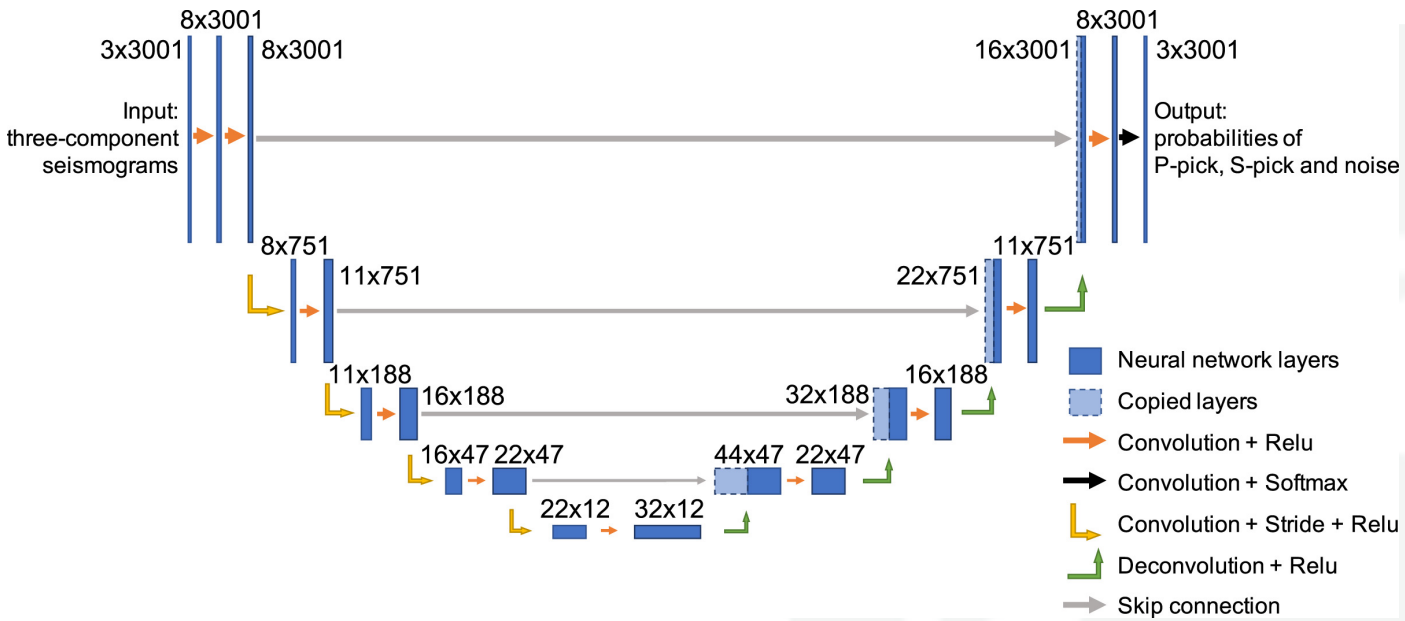


Shiina et al. 2013

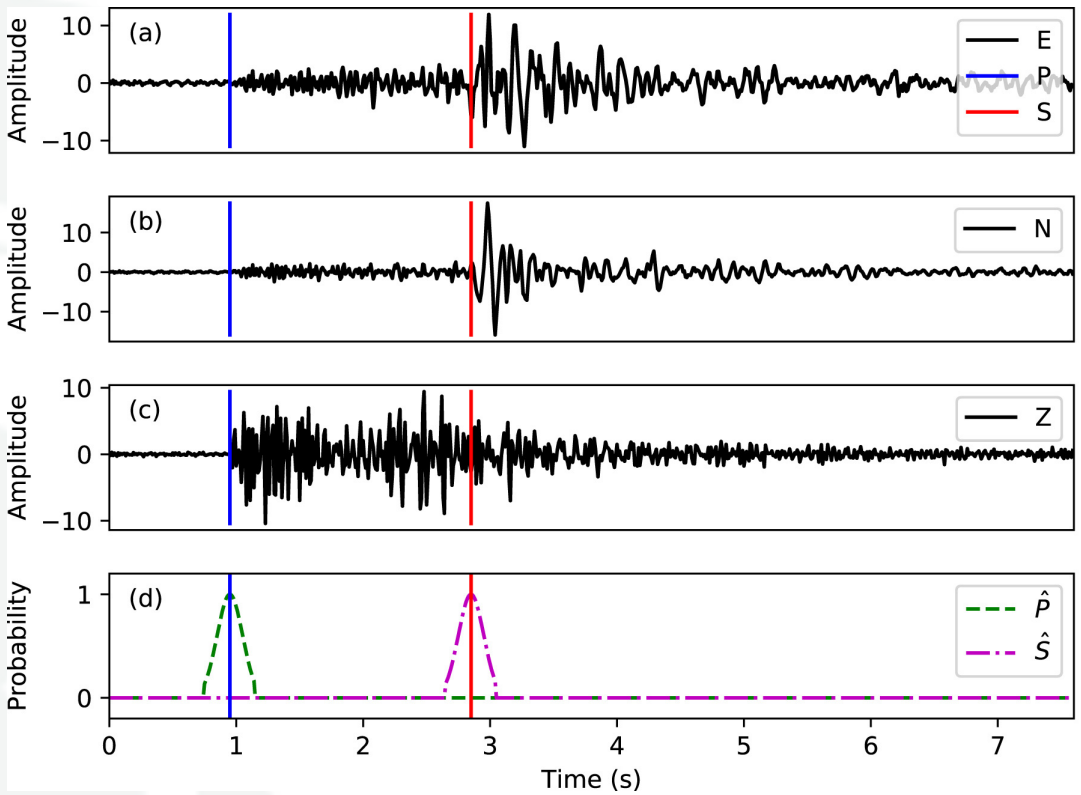
- Ps waves provide very good constrain of the upper slab interface
- We estimate the slab crust can be ~10km thick
- Intermediate-depth earthquakes might be related to dehydration (both in the slab crust and the slab mantle)

To get a more accurate slab upper interface, we need more data, which humans can not pick them all with the same standard!

Previous work: PhaseNet

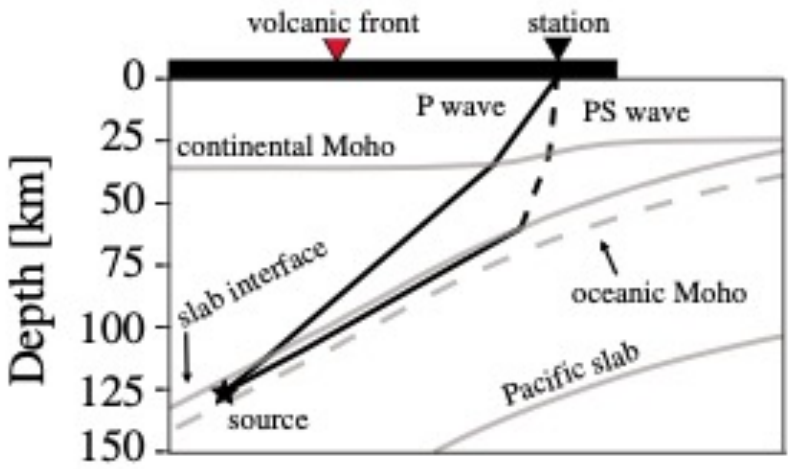


Zhu and Beroza, GJI, 2018

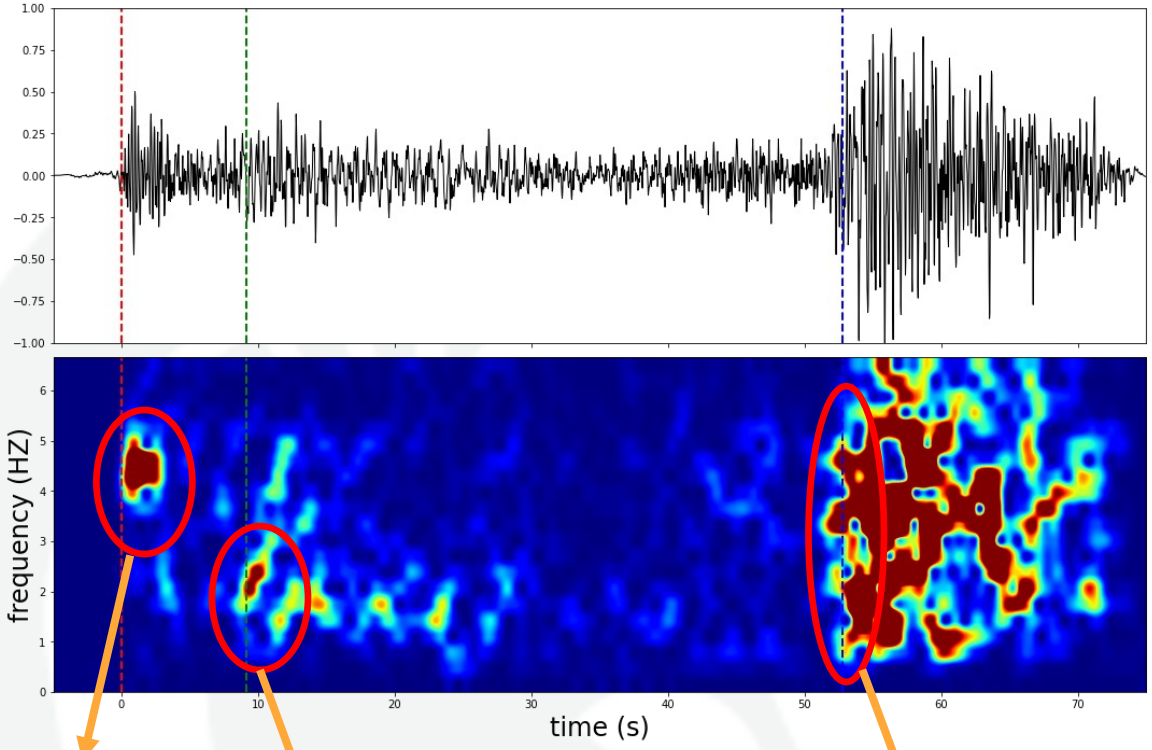


1. PhaseNet has high accuracy in detecting P and S arrivals.
2. Converted phases are usually weak and hard to detect.

From seismology to machine learning



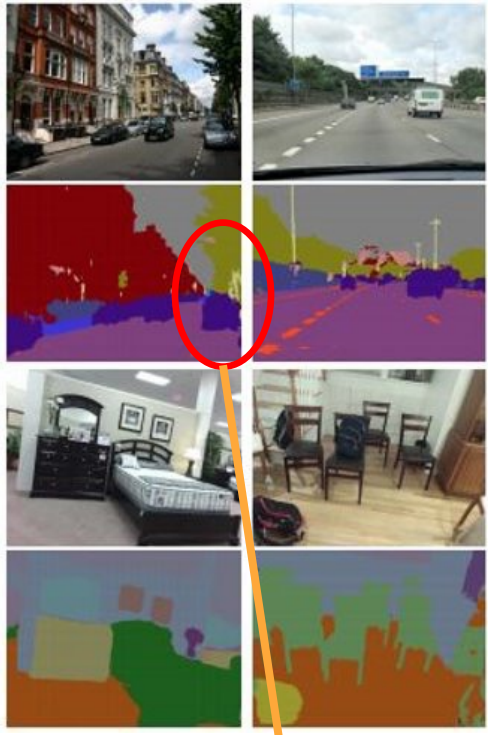
Shiina et al. 2013



P wave

PS wave

S wave

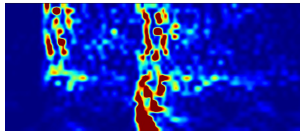


The car in front of us

There is a great similarity between seismology with autonomous driving!

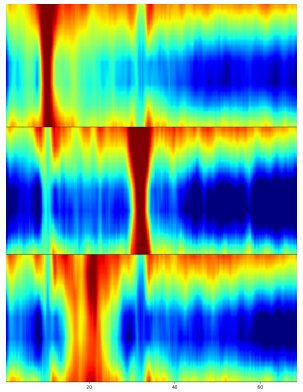
What is the method here? Semantic Image segmentation (image classification at the pixel level).

PhaseNet-TF model architecture



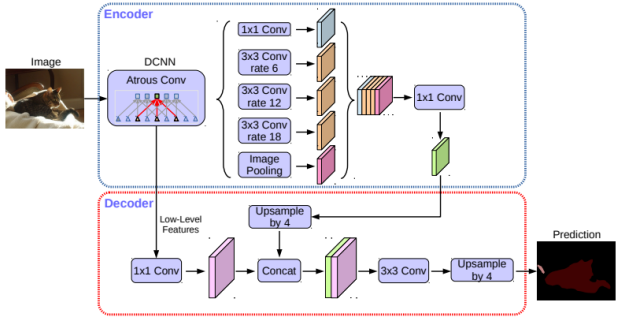
spectrogram

semantic image segmentation

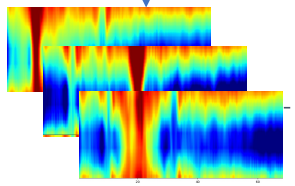


Output from the semantic image segmentation layer

DeepLabV3+ with Resnet34 as the encoder backbone

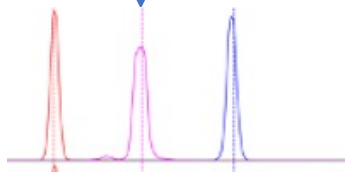


Model architecture figure cited from Chen et al. ECCV 2018



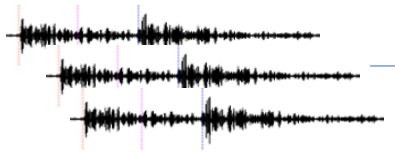
Linear

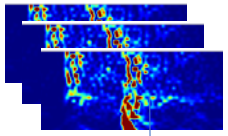
Softmax



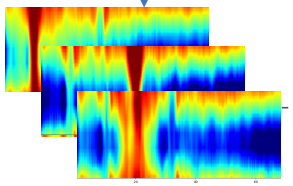
Possibility prediction for phases
Here P, PS, S in order

PhaseNet-TF model architecture



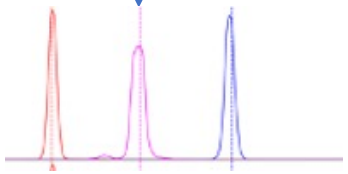


semantic image segmentation



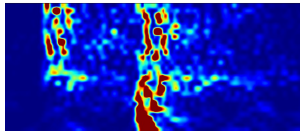
Linear

Softmax

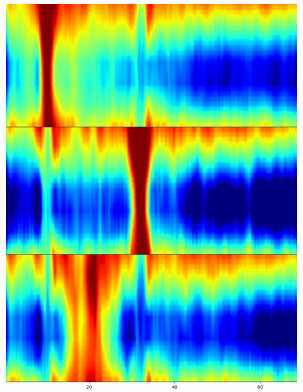


Possibility prediction for phases
Here P, PS, S in order

(2)

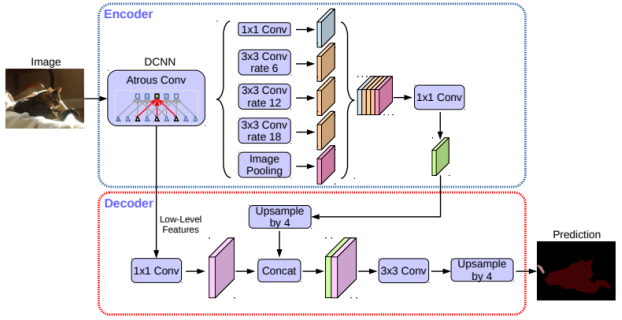


spectrogram



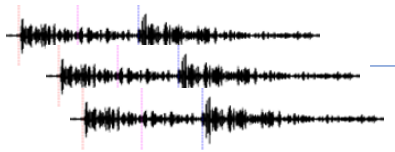
Output from the semantic image segmentation layer

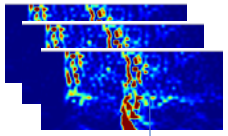
DeepLabV3+ with Resnet34 as the encoder backbone

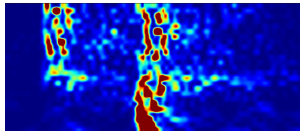


Model architecture figure cited from Chen et al. ECCV 2018

PhaseNet-TF model architecture

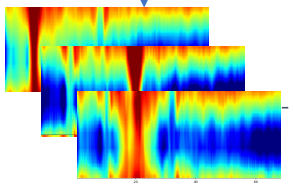


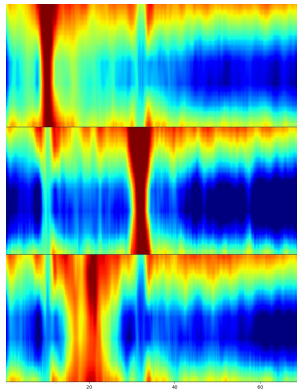




spectrogram

semantic image segmentation

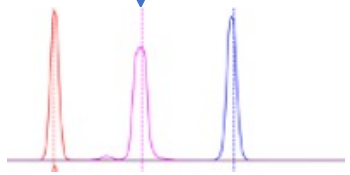




Output from the semantic image segmentation layer

Linear

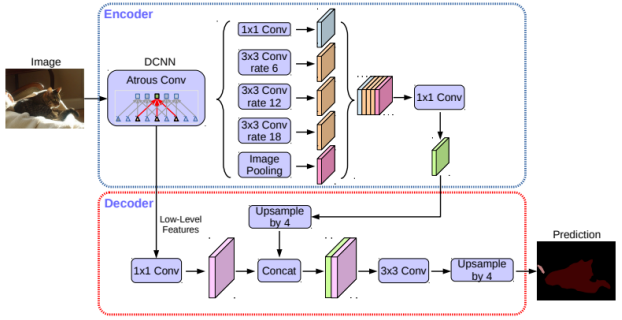
Softmax



Possibility prediction for phases
Here P, PS, S in order

(3)

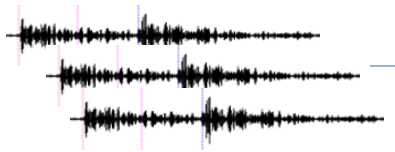
DeepLabV3+ with Resnet34 as the encoder backbone

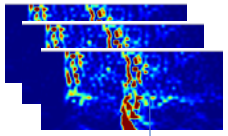


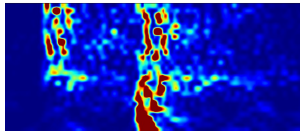
Model architecture figure cited from Chen et al. ECCV 2018

PhaseNet-TF model architecture

(1)

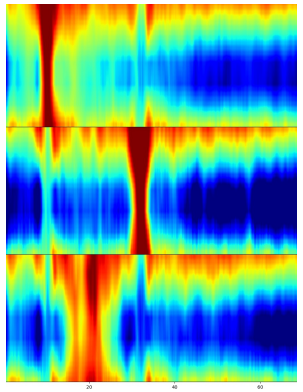




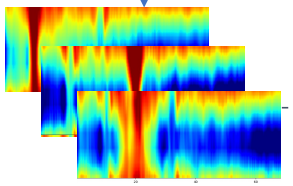


spectrogram

semantic image segmentation

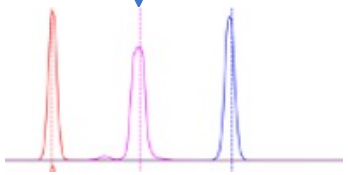


Output from the semantic image segmentation layer



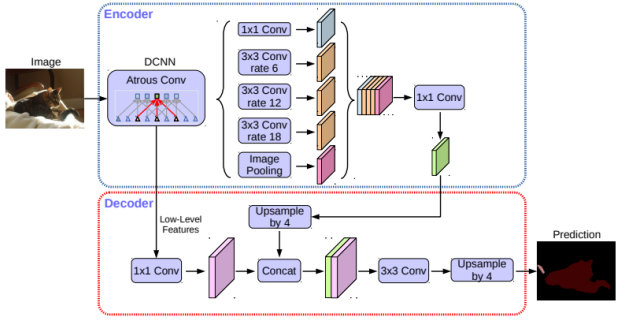
Linear

Softmax



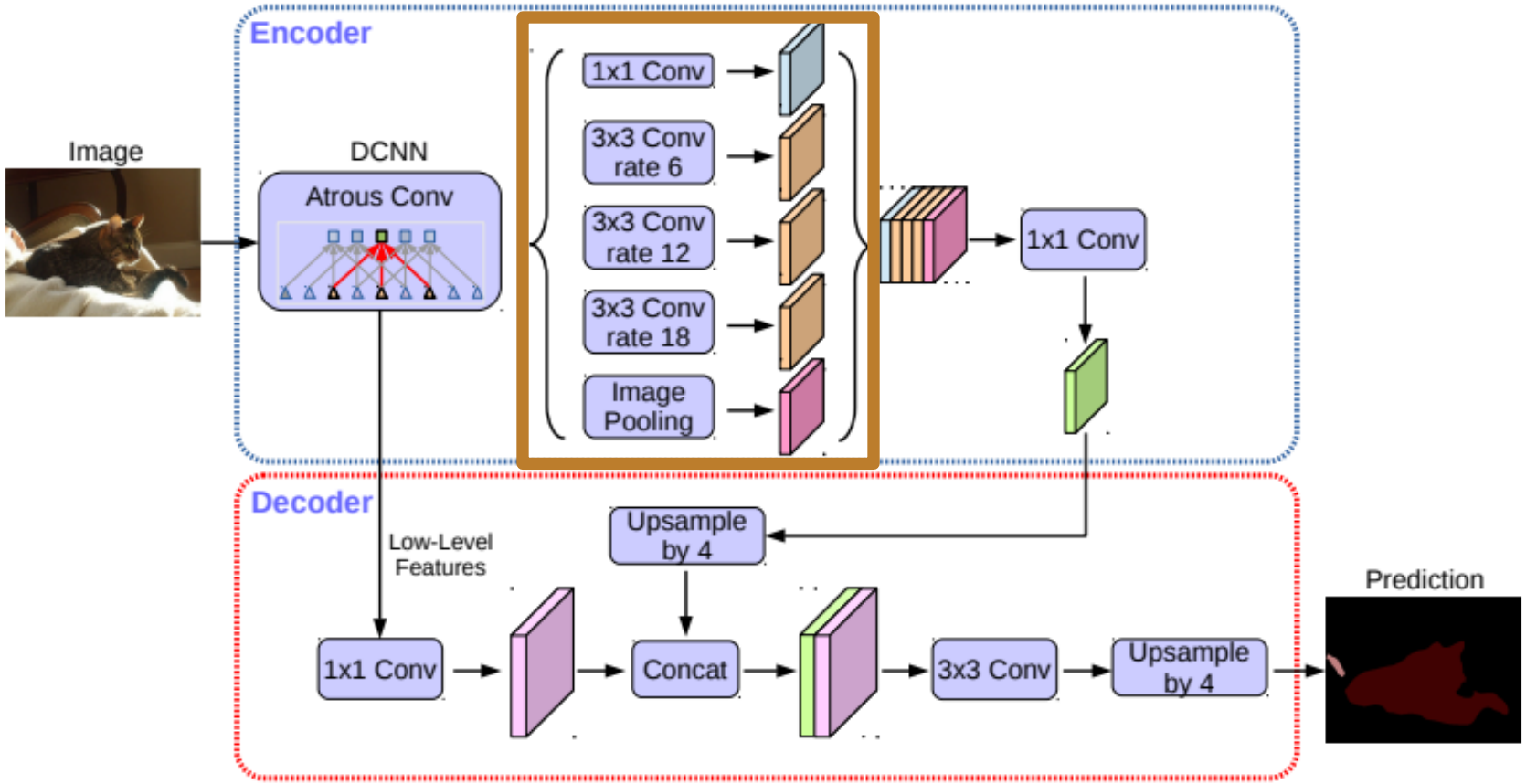
Possibility prediction for phases
Here P, PS, S in order

DeepLabV3+ with Resnet34 as the encoder backbone

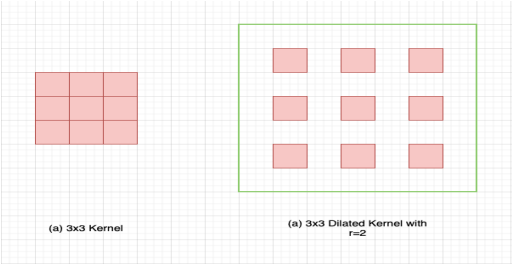


Model architecture figure cited from Chen et al. ECCV 2018

DeepLabV3+, one of the best image segmentation models



1. Atrous conv layer

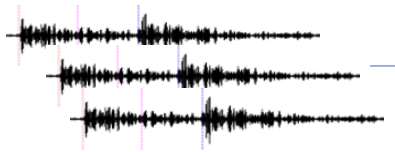


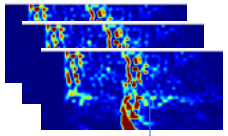
2. Use ASPP (in the brown box, Atrous Spatial Pyramid Pooling) to improve receptive field at different scales

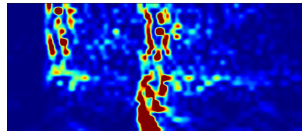
3. The skip layer between for encoder and decoder for the shallow layer's feature fusion

Model architecture figure cited from Chen et al. ECCV 2018

PhaseNet-TF model architecture



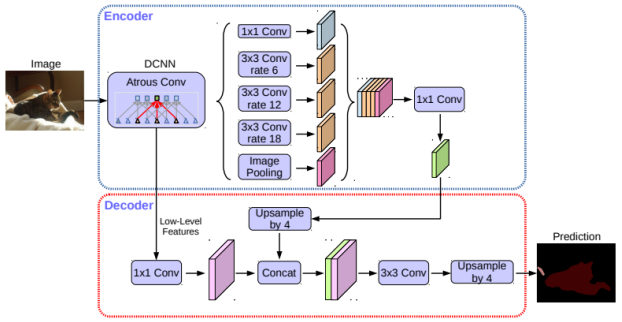




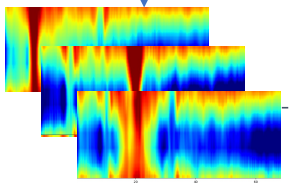
spectrogram

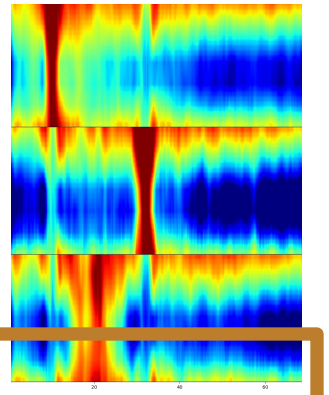
semantic image segmentation

DeepLabV3+ with Resnet34 as the encoder backbone



Model architecture figure cited from Chen et al. ECCV 2018



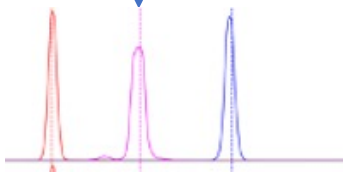


Output from the semantic image segmentation layer

Linear

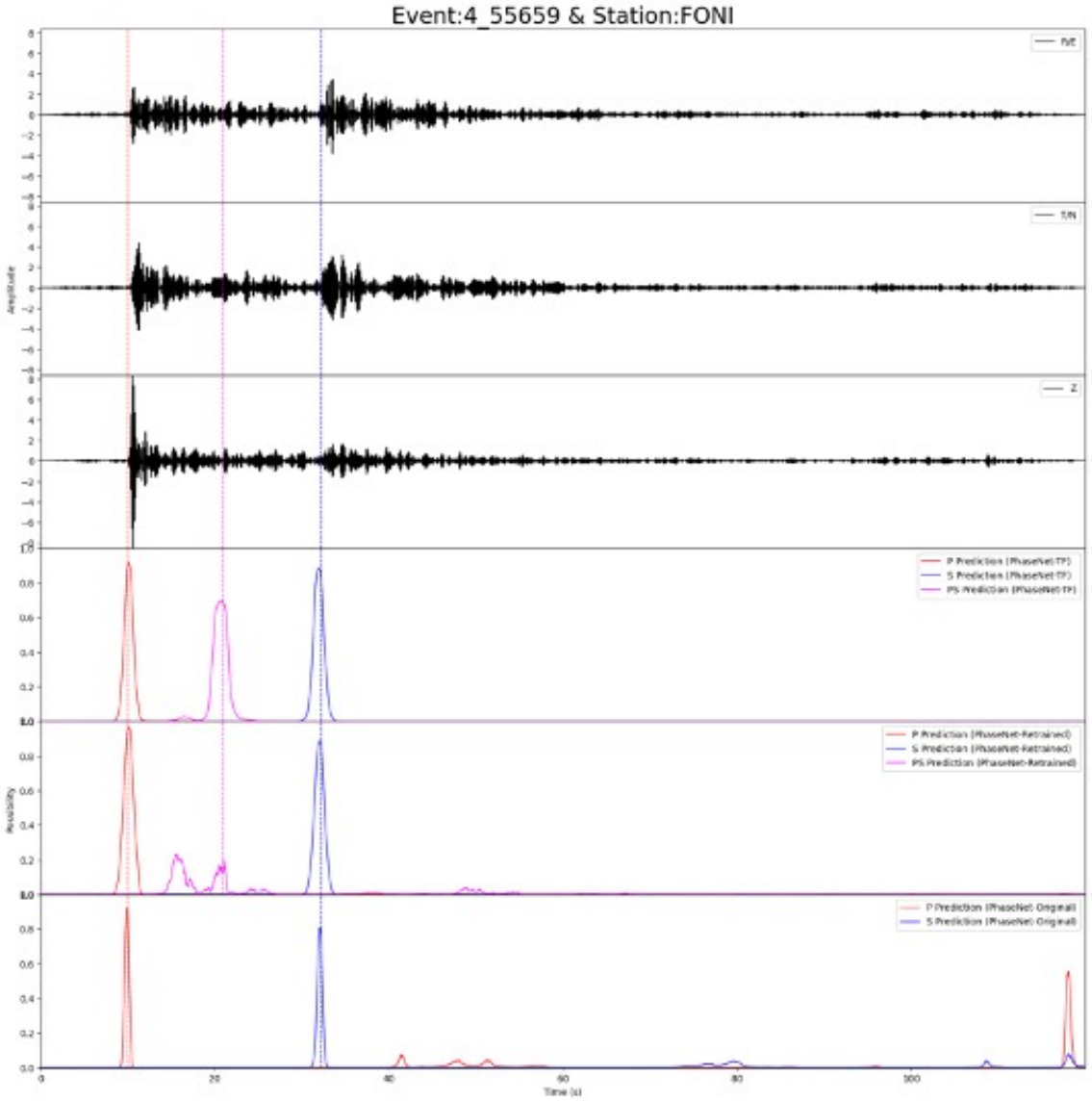
Softmax

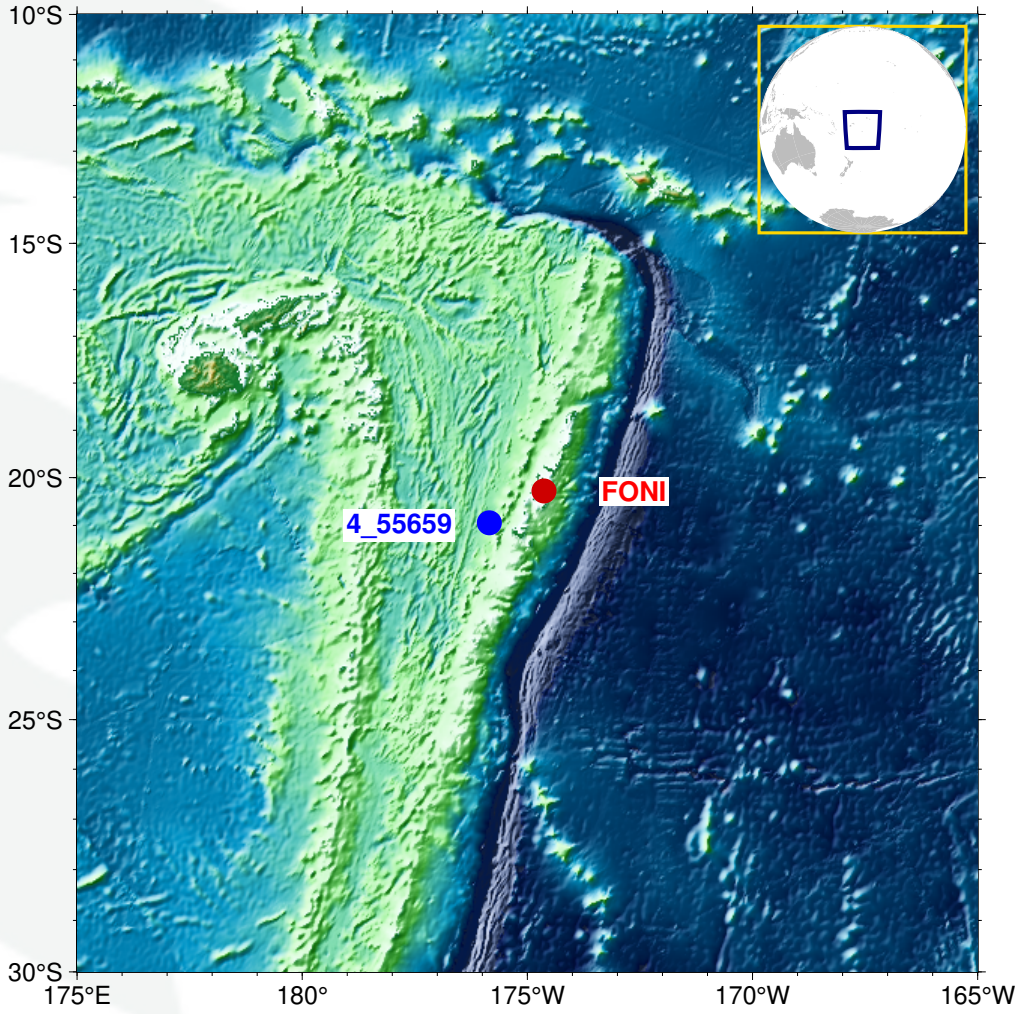
(4)



Possibility prediction for phases
Here P, PS, S in order

One example, here only PhaseNet-TF can detect weak PS



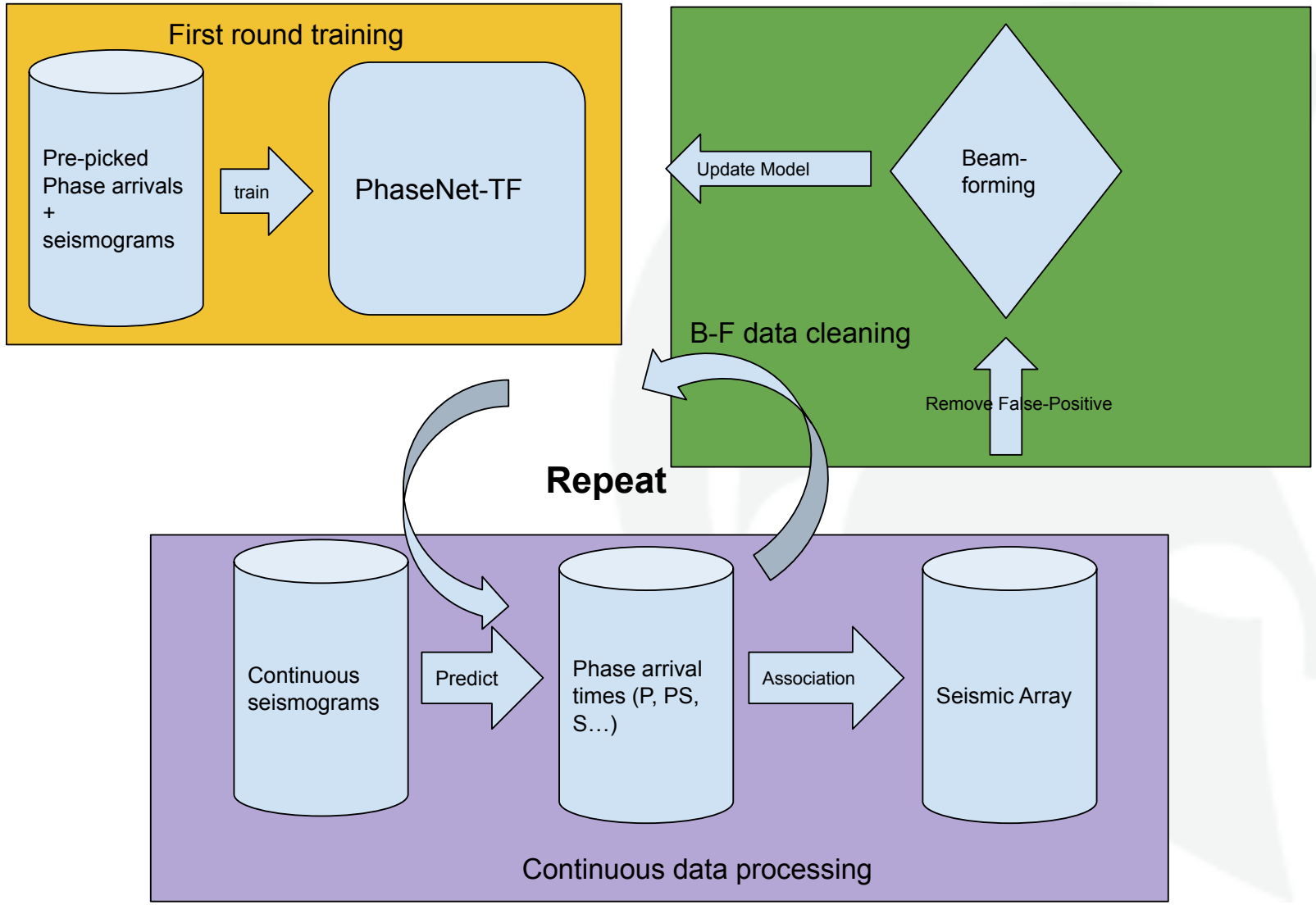


PhaseNet-TF is better in detecting more PS phases (Recall)

$True: \lvert t_{predicted} - t_{labeled} \rvert < 0.5\ s$	Precision	Recall	F1
PhaseNet-TF (our work)	P: 0.8704 S: 0.7285 PS: 0.4803	P: 0.7626 S: 0.5848 PS: 0.4338	P: 0.8129 S: 0.6488 PS: 0.4558
PhaseNet - retrained	P: 0.8774 S: 0.7403 PS: 0.4585	P: 0.7773 S: 0.5737 PS: 0.2742	P: 0.8243 S: 0.6464 PS: 0.3432
PhaseNet – originally trained model from paper	P: 0.7948 S: 0.6185 PS: N/A	P: 0.5963 S: 0.4239 PS: N/A	P: 0.6814 S: 0.5031 PS: N/A

1. It's reasonable that retrained PhaseNet has better performance after training with the Tonga dataset.
2. PhaseNet-TF and PhaseNet have comparable performance in detecting P and S.
3. PhaseNet-TF has a higher recall value in detecting PS, thus better detecting weak PS signals.

Repeating training with help of the array seismology



Workflow Summary:

1. Phase-picker: PhaseNet-TF
2. Catalog builder: Phase association (e.g. Gamma)
3. Prediction cleaner: use array seismology methods such as beam-forming + jackknife method to remove false-positive predictions

Summary

- **We develop a deep learning algorithm that can detect body waves, including P, S, and weak converted phases such as PS from the frequency domain.**

- **With the help of the semantic image segmentation techniques, the model has shown the advantage of improving the detection rate of the converted phases (Recall).**

- **We propose a workflow combing PhaseNet-TF, phase association, and beam-forming to refine the deep learning model and build the PS arrival time catalog in Tonga.**

- **We expect this package to increase the number of PS-wave observations dramatically and help improve the current slab structure models. In addition, this work can help us better understand the Earth's water cycle from the subduction process.**